	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		EXPERIMENT 05
Branch	**CSE, CS&IT, Data Science, AI&ML, IoT, Cyber Security**	Programme	**B.Tech**
Course Name	**Algorithm Analysis and Design - 2**	Semester	**4th**
Course Code	**CSE 2632**	Academic Year	**2025-2026**
Learning Level (LL)	**L1**: Remembering	**L3:** Application	**L5**: Evaluation
	L2: Understanding	**L4**: Analysis	**L6**: Creation

Each Question Carries 5 Marks

Objective of Experiment: Constructing the minimum weight spanning tree of the given weighted-undirected graph. Estimating the shortest path from a given source to all other vertices for the given weighted graph.

Q's	Questions	COs	LL
1	Write a program to find the minimum weight spanning tree of the given weighted-undirected graph using the PRIMS algorithm.		**L3**
2	Write a program to find the minimum weight spanning tree of the given weighted-undirected graph using the KRUSKAL.		**L3**
3	Write a program to estimate the shortest path from a given source to all other vertices in the given weighted graph using the Dijkstra Algorithm.		**L3**

Course Outcomes	CO1	understand the network flow problem and apply it to real-world problems
	CO2	- Distinguish between computationally tractable and intractable problems. - Define and relate class-P, class-NP and class NP-complete, PSPACE, and PSPACE-complete. - Given a problem in NP, define an appropriate certificate and the verification algorithm.
	CO3	Understand approximation algorithms and apply this concept to solve problems.
	CO4	Understand local search techniques and apply this concept to solve problems.
	CO5	Understand randomization and apply this concept to solve problems.
	CO6	Identify and apply an appropriate algorithmic approach to solve a problem and explain the challenges to solve it.